Adaptive S3 rows: finite degree and relation certificates

Quadratic-twist study, PR2381

25 October 2026; adaptive-s3-20261025

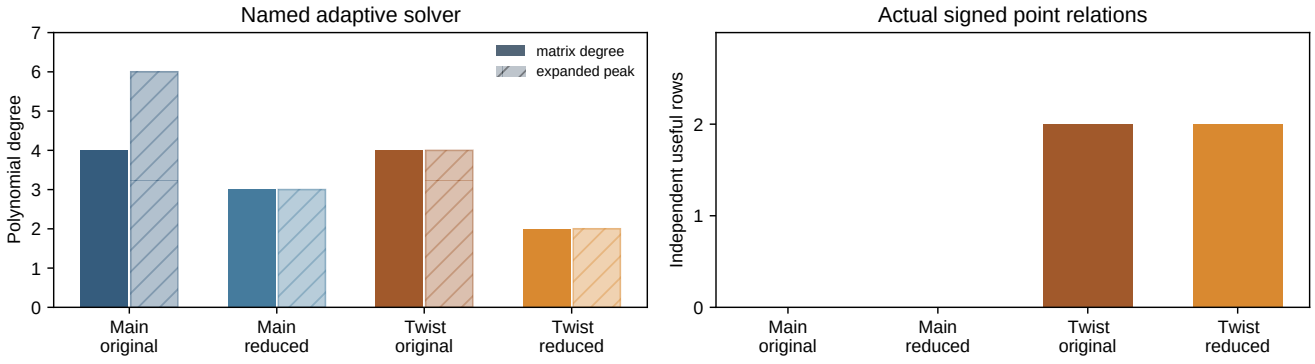
Executed scope. The preregistered adaptive row solver ran on all 98 size-two cases, 188 main/twist base comparisons and 1,384 target comparisons. Independent C++ and Rust agree on every polynomial witness, matrix rank, promoted row, actual elliptic-point lift, useful coefficient rank, stopping degree and scalar recovery. Reduced inputs have 24 target comparisons with both higher useful rank and lower operational matrix degree; original S3 inputs have zero. These counts are exact finite observations, not probabilities, timings or claims of intrinsic regularity.

Actual F43 fixture

Use $E : y^2 = x^3 + x$, $T : Y^2 = X^3 + 4X$ over F_{43}, order-11 generators $P = (31,18)$, $U = (24,26)$ and factor-base representatives $(4,5),(13,19)$ and $(15,9),(17,6)$. These are existing model identities `icv1-fp6-t0-74779950` and `icv1-fp6-t0-c6a8a47c`. For their first targets:

Audited quantity	Main raw	Main reduced	Twist raw	Twist reduced
Useful independent rows	0	0	2	2
Constrained ordered roots	0	0	3	3
Solver matrix degree	4	3	4	2
Largest expanded witness degree	6	3	4	2
Terminal witness degree	6	3	4	2

F43 / order 11, first target; adaptive-s3-20261025



98 cases and 1,384 target comparisons audited; bars show one declared fixture

Values are stored in `coverage-data.json`; the plotting source only lays out those values. The figure shows one fixture within the full audited population.

What the solver certified

The actual reduced inputs are $h_m = 41xy + 39(x+y) + 27$, $h_t = 17xy + 3(x+y) + 41 = 17(x-15)(y-15)$, with their factor-base membership quadratics. Each promoted row carries three exact multiplier polynomials in the supplied inputs. The named solver builds all admissible multiples of the pool through degree D, computes a full row space, reinserts a new lower degree row only when absent from its prior lower-degree space, and rebuilds to a fixed point. Positive acceptance requires every input active and complete interpolation of all actual roots. Expanded witness peak includes generated rows before reduction. Caps are explicit and none was reached.

The main reduced D=2 input space excludes 1. At D=3 the checked identity is

$$(37+16x+16y)h_m+(21+32y)(x^2+26x+9)+(21+32x)(y^2+26y+9)=1 \pmod{43}.$$

The twist has exactly the roots $(15,15),(15,17),(17,15)$; every quadratic vanishing at them is in the span of its three input quadratics. Thus the positive D=2 certificate and negative main D=3 certificate concern the actual equations, rather than abstract row counts.

Source DLP and mathematical scope

The previously proved point map $\phi(O)=O$, $\phi(x,y)=(41x,11y)$ is an injective additive map from this main curve to this twist, sending P to U and preserving integer scalars. Here it is a base-field isomorphism. The new native routine takes each of 11 separate main points, derives the twist base logs $5,7$ from actual generator relations, queries the certified reduced solver on recorded target shifts, and verifies the recovered source scalar $0,\ldots,10$ on E. The identity has no solver attempt; every nonidentity attempt retains its full certificate. The solver never receives the known-answer label.

The original-S3 main row space terminates at matrix degree 4, but its expanded input witness reaches degree 6. This is lower *operational* matrix degree under reinsertion, not a degree-4 original-input refutation. Both reduced arms have the same previously proved homogeneous input cutoff 2. No intrinsic regularity separation, F4/F5 timing, asymptotic advantage or complete ECDLP speedup is established. Different factor bases account for the useful-rank difference in this fixture; transported-base controls identify the relevant coordinate equivalence.

Checks and next test

The new Lean project proves row-adjunction ideal invariance, the main D=2 dual obstruction, the D=3 unit, exact twist roots and full quadratic vanishing space. The universal cleared unit polynomial has source-bound identity certificate `IDC1h9eea191ebc3a8def`, checked at 32+16 native points; false identity and source binding reject. Independent Rust checks all native rows and recoveries. Five rehashed semantic corruptions (witness, degree, target coverage, source and scalar) reject without an audit receipt. The unmodified scoped proof gate and mandatory library suite were run for this round; the exact receipts and raw failures are retained under `validation/adaptive-s3-20261025`.

The next discriminating test uses non-isomorphic twist pairs and source-matched bases, then charges factor-base setup, relation generation, row-space solve, point transfer and target verification. A claim of intrinsic regularity would need the same specified homogenization and term order on both actual ideals. General-model normalization, exceptional S3/S4 lifting, generic pairing, trace-zero and coordination obligations remain separate.